

Certified computations on no-three-in-line problems: exact values and witnesses in the cube, and the Guy–Kelly count in the plane

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Abstract

We report certified computations on four no-three-in-line questions, two in the cube and two in the plane, by one method: exact decision procedures inside symmetry strata, every witness re-verified by a second program sharing no code, every number traced to a public journal. (I) Let $a(n)$ be the largest number of points of $\{0, \dots, n-1\}^3$ with no three collinear (A399138). We determine $a(1), \dots, a(6) = 1, 8, 16, 28, 40, 64$ with DRAT-certified unsatisfiability proofs, give certified lower bounds $a(7) \geq 73$, $a(8) \geq 94$, $a(9) \geq 116$, $a(10) \geq 138$, $a(11) \geq 164$, and prove $a(p) \geq p^2$ for every prime p . The optima share a layer structure $2n^2 - 2n + 4$ that provably fails at $n = 5$ and $n = 7$. (II) For $b(n)$, the largest number with no four coplanar (A280537), nineteen public configurations give lower bounds for $9 \leq n \leq 29$, four of which improve the known bounds and their monotone closure ($b(12) \geq 31$, $b(21) \geq 47$, $b(22) \geq 49$, $b(27) \geq 56$); the cyclically invariant subspace has maximum 23 at $n = 9$ and 26 at $n = 10$; four kinds of symmetry are incompatible with the problem; no upper bound beyond $3n$ is known. (III) The Guy–Kelly first-moment heuristic for the plane is audited against exact counts (A000755 to $n = 20$): its corrected constant comes out in closed form, $\pi/\sqrt{3}$, the threshold crossing $n = 493$ is reproduced, and its error is shown to depend on the shape of the question rather than on n alone, with an unbounded multiplier; whether the residual error is $\Theta(n)$ or $\Theta(n \ln n)$, which decides whether the constant survives, cannot be told by counting, and we measure how far from telling we are. (IV) The direction spectrum of the $2n$ -point solutions is measured, and a line model with no fitted parameter reproduces its shape (fifteen constants within 12%, seven predicted blind) but not its scale. Withdrawn claims are kept in the text; each part states what it does not establish.

1 Introduction

The no-three-in-line problem asks for the largest subset of the $n \times n$ grid with no three points collinear. Two points per row bound the answer by $2n$; Flammenkamp’s database records $2n$ -point solutions for sizes up to $n = 57$ [2], and the heuristic of Guy and Kelly [1] predicts that for large n only about $1.814n$ points can be placed. The problem has two natural analogues in the cube $\{0, \dots, n-1\}^3$: forbid three collinear points, where Pór and Wood [13] proved that the maximum is $\Theta(n^2)$ but no exact values had been recorded; or forbid four coplanar points (OEIS A280537), where the known terms stop at $n = 8$ and only lower bounds are known beyond. For the history and state of the planar problem we refer to the survey [4] and to Prellberg’s constraint-programming computations [5]; the prior art specific to each part is stated inside it (Sections 4.1, 5.1 and 6.5).

This article reports computations on four questions in this family, two in the cube and two in the plane, by one method. Every extremal value is obtained by an exact decision procedure

— a propositional encoding whose unsatisfiability is certified, or an exhaustive traversal whose completeness is checked — and every lower bound is a witness configuration re-verified by a second program that shares no code with the search, after that program has been shown to reject a deliberately corrupted witness. Searches for witnesses are run inside the strata of a symmetry group, where the exact maximum of a stratum is often within reach; a stratum maximum is a lower bound on the true maximum and is reported as such. Estimates are tested where exact values exist, and their reach is measured before their extrapolations are believed. Every number in the text traces to a journal file in the public repository (Section 8), and the claims that earlier versions of the parts had to withdraw are kept in the text rather than erased.

The results, part by part.

Part I (Section 3). For $a(n)$, the largest number of points of $[n]^3$ with no three collinear: $a(1), \dots, a(6) = 1, 8, 16, 28, 40, 64$ (the sequence was not in the OEIS when this work began and is now A399138); certified lower bounds $a(7) \geq 73$, $a(8) \geq 94$, $a(9) \geq 116$, $a(10) \geq 138$, $a(11) \geq 164$; the bound $a(p) \geq p^2$ for every prime p ; and a layer structure $2n^2 - 2n + 4$ of the optima that holds for $n = 2, 3, 4, 6$ and provably fails at $n = 5$ and $n = 7$.

Part I' (Section 4). For $b(n)$, the largest number with no four coplanar: nineteen public configurations for $9 \leq n \leq 29$, four of which improve the bounds known from a 2017 OEIS comment and a 2016 programming contest together with their monotone closure ($b(12) \geq 31$, $b(21) \geq 47$, $b(22) \geq 49$, $b(27) \geq 56$); the exact maxima 23 and 26 of the cyclically invariant subspace at $n = 9$ and $n = 10$; and the structure of the symmetry strata, including four kinds of symmetry that the problem excludes outright.

Part II (Section 5). The Guy–Kelly heuristic, a first-moment estimate, is audited on the quantity it predicts — the number of $2n$ -point solutions, known exactly to $n = 20$ — rather than at its threshold. Its corrected constant $\pi/\sqrt{3}$ is derived in closed form from the asymptotics of the number of collinear triples; its error is measured as a function of the shape of the question and found to be an unbounded multiplier; and the question whether that error is $\Theta(n)$ or $\Theta(n \ln n)$ is shown to be undecidable by the kind of computation available.

Part II' (Section 6). On the same database of solutions, the mean number of point pairs per direction is measured, and a line model with one theorem and no fitted parameter reproduces the shape of the spectrum, including seven constants predicted before they were measured, while its scale has to be imposed.

Section 7 joins the parts: a first-moment threshold in the cube that the witnesses of Part I' crossed, and the Guy–Kelly form of the estimate tested on the cube enumerations. Section 8 states the verification protocol common to all parts and lists the public records; Section 9 discloses how the work was produced. The four parts were first circulated as separate notes (their records are listed in Section 8); this article consolidates them into one account with one notation, one section on verification and one disclosure, and supersedes them.

2 Notation and public objects

Throughout, $[n] = \{0, 1, \dots, n-1\}$; the grids are $[n]^2$ and $[n]^3$. A line is *rich* in a grid if it contains at least three of its lattice points, and a plane is *rich* in $[n]^3$ if it contains at least four. A line carrying at most two lattice points forbids nothing, and a plane carrying at most three forbids nothing; hence

$$S \subseteq [n]^3 \text{ has no three collinear points} \iff |S \cap L| \leq 2 \text{ for every rich line } L, \quad (1)$$

and likewise S has no four coplanar points iff $|S \cap P| \leq 3$ for every rich plane P . We call a set *admissible* for the prohibition at hand when it satisfies it. Both conditions are propositional formulae on n^3 variables, one cardinality constraint per rich line or plane; this reformulation is used throughout Parts I and I'. In the no-four-coplanar problem three collinear points are forbidden as well, since with any fourth point they are coplanar.

The three extremal quantities of this article are

$$\begin{aligned} a(n) &= \max\{|S| : S \subseteq [n]^3, \text{ no three of } S \text{ collinear}\} \quad (\text{OEIS A399138}), \\ b(n) &= \max\{|S| : S \subseteq [n]^3, \text{ no four of } S \text{ coplanar}\} \quad (\text{OEIS A280537}), \end{aligned}$$

and, in the plane, the maximum $2n$ itself: a *solution* is a set of $2n$ points of $[n]^2$ with no three collinear, the maximum possible since every row carries at most two points. We write $s(n)$ for the number of solutions (OEIS A000755), $T(n)$ for the number of collinear triples of $[n]^2$, and D_4 for the dihedral group of the square (order 8). The symmetry group of the cube has order 48 and 33 conjugacy classes of subgroups; for a subgroup H the H -*stratum* of a problem is the family of H -invariant admissible sets, and its maximum is a lower bound for the maximum of the problem — the exact maximum of the stratum when the optimisation closes, a lower bound inside the stratum otherwise. Lexicographic leader constraints for the cube group, the strata of its subgroups and the D_4 -classes of planar solutions are used in Parts I, I' and II' respectively.

Two elementary bounds. Each of the $3n^2$ axis-parallel lines of $[n]^3$ carries at most two points of an admissible set and each point lies on three of them, so $a(n) \leq 2n^2$; this is the trivial upper bound, attained at $n = 2$. And:

Proposition 2.1. $b(n) \leq 3n$ for every n .

Proof. Each of the n planes $x = \text{const}$ carries at most three points of an admissible set. □

This is the only upper bound for $b(n)$ above $n = 8$ that we can prove.

Public objects. The OEIS entries produced or extended by this work: A399138 (new: $a(1), \dots, a(6)$, approved August 2026), A280537 (edit: $b(12) \geq 31$ and $b(n) \leq 3n$, approved September 2026) and A000755 ($s(20) = 941\,580$, approved August 2026). Programs, journals, witnesses, manifests and the verification protocol are public in the repository <https://github.com/iwasborninbali/saturation>; the configurations of Part I are archived separately [20]; the data of Parts II and II' are Flammenkamp's [2]. Section 8 lists the records and the ancillary files.

3 Part I: the cube, no three collinear

3.1 Method: a certified decision procedure

By (1), the question “is there an admissible set of size M in $[n]^3$?” is a Boolean satisfiability problem with n^3 variables — 125 for $n = 5$, 216 for $n = 6$ — one at-most-two cardinality constraint per rich line, and one global at-least- M constraint. The global counter is a totalizer truncated at $2n^2$; the truncation is legitimate because each of the n^2 axis-parallel lines carries at most two points, so $a(n) \leq 2n^2$. We add the 47 lexicographic leader constraints of the cube group, which preserve satisfiability because the property in (1) is invariant under that group.

We adopted this route after measuring the alternative. A prefix-split exhaustive search on 42 cores closed 5 of 204 prefixes in two hours at $n = 5$, extrapolating to roughly 88 core-days; the same question is decided by a SAT solver in minutes. Speed, however, is not the reason. An exhaustive search establishes completeness by *its own* covering argument, which only its authors' code can check; an unsatisfiability answer comes with a DRAT certificate that an independent, widely used checker verifies without trusting our solver or our code at all.

Three separate links. It is worth being explicit about what is and is not certified, because the three claims below rest on different foundations.

1. *The formula expresses the problem.* Verified two independent ways: by comparing the set of triples forbidden by the rich-line constraints against the set with vanishing cross product, and by forcing each collinear triple as an assumption and requiring unsatisfiability. At $n = 4$ and $n = 5$ there are 376 and 1858 collinear triples; none was left unforbidden and none was forbidden spuriously. A DRAT checker cannot verify this link — it knows nothing of cubes and lines — and this is exactly where an encoding bug would hide.
2. *The formula is unsatisfiable.* Verified by `drat-trim` [15] on the certificate emitted by `kissat` [14].
3. *A witness is admissible.* Verified by exhaustive integer arithmetic over all $\binom{|S|}{3}$ triples, by a program sharing no code with the search.

Links 1 and 2 give the upper bound; links 1 and 3 give the lower bound.

Removing link 1 for small n . The two checks above verify that the encoding is faithful. A stronger thing is possible: to compute the same values without an encoding at all, so that link 1 does not need to be trusted because it is not used. We wrote a plain exhaustive traversal that places points in the cube and tests triples by integer cross product — the literal definition — with no propositional formula, no solver and no certificate. Two points kill the line through them, so every cell carries a counter of dead lines covering it and a cell is available while that counter is zero; a branch is abandoned when the points placed plus the cells still available cannot exceed the best found. Run from a bar of zero, so that no value can be fed in and echoed back:

n	result	nodes
1	1 (exhausted)	2
2	8 (exhausted)	37
3	16 (exhausted)	49 812

These agree with the certified values and share nothing with the certified route but the definition of the prohibition itself. Our own full enumeration at $n = 3$ visits all 3 813 884 admissible subsets (the empty one included) and finds exactly *two* optima of size 16 — a chiral pair, each with stabiliser of order 24 in the full cube group, both with layer profile $[6, 4, 6]$ along all three axes. At $n = 4$ a traversal from a bar of 27 *exhausted* its tree (9 012 859 649 nodes, 3180 s), proving $a(4) = 28$ with no encoding at all — a bar is harmless under exhaustion, since it prunes only branches unable to exceed it. The count at $n = 4$ is now ours as well: a 32-shard traversal (sharded by first cell, run to completion on every shard, 13 215 017 061 nodes in all) finds exactly 14 optima of size 28, in three classes under the full cube group with stabiliser orders 8, 8, 24 ($6 + 6 + 2 = 14$) — in agreement with an independent external recount. Every optimum at $n \leq 4$ therefore has the layer profile $[2n, (2n - 2)^{n-2}, 2n]$: the empirical observation of Section 3.3 is a fact up to $n = 4$.

Correction. An earlier version of this paragraph quoted two numbers received from the first solver’s message and never passed through the verification bench of this very part: “2 220 075 subsets” and “the count 10 960”. Both are withdrawn, and both origins have since been reconstructed: 2 220 075 is exactly $\binom{27}{8}$ — the enumeration of all 8-subsets of 3^3 that verifies the uniqueness of the 8-point optimum of the *no-four-coplanar* problem (A280538 gives 1 there) — and $10\,960 = 48 \cdot 232 - 176$ is the number of labelled optima of that same neighbouring problem at $n = 4$, where A280538 gives 232 classes. Two numbers from the wrong problem, quoted into the section about removing trust; recorded, not smoothed.

We are explicit about how far this extends. From a bar of zero the traversal reaches 28 at $n = 4$ without exhausting in an hour ($8.85 \cdot 10^9$ nodes); from a bar of 27 it exhausts ($9.01 \cdot 10^9$ nodes, 53 minutes), so $a(4) = 28$ is established twice over as well. The honest division: for

$n \leq 4$ both the value and the bound are encoding-free in addition to certified; for $n \geq 5$ the encoding is checked but used.

Calibration. Before computing anything new we recomputed, as a calibration, the two smallest nontrivial values $a(3) = 16$ and $a(4) = 28$ — there is no prior published source for them that we could find, so the calibration is against our own earlier runs, not against the literature — each with unsatisfiability at $M + 1$ in under a second, independently on two machines. For instances that resist a single solver call we split into independent pieces by fixing the contents of the first columns; the split is complete because three points of a column are collinear, so subsets of size at most two exhaust it. The aggregator refuses a verdict unless every piece of the manifest is present and unsatisfiable — a missing piece and an unsatisfiable one must never look alike.

3.2 Exact values

n	$a(n)$	$a(n)/n^2$	how established
1	1	—	trivial
2	8	2.00	exhaustive; meets the trivial bound $2n^2$
3	16	1.78	exhaustive, and certified
4	28	1.75	exhaustive, and certified
5	40	1.60	certified
6	64	1.78	certified

The value $a(6) = 64$ appears to be new: when this work began, a search of the OEIS for 1, 8, 16, 28, 40 and for 8, 16, 28, 40, 64 returned nothing, and the sequence itself was not in the database (it is now A399138).

For $n \leq 4$ the values were also produced by two independent exhaustive programs whose node counts differ by a factor of about 72 ($17/3855/4.21 \cdot 10^7$ against $17/20068/3.03 \cdot 10^9$ for $n = 2, 3, 4$), so their agreement is not an artefact of a shared design.

For $n = 5$ and $n = 6$ the decisive question — does a set of 41, respectively 65, points exist? — was split into independent pieces by fixing the contents of the first two z -columns, and the heaviest pieces were split once more on the third column. Both splits are complete because three points of a column are collinear, so subsets of size at most two exhaust it. The final tally, reproduced in `logs/no3_3d/FINAL_VERDICT.txt`:

	$n = 5, M = 41$	$n = 6, M = 65$
top-level pieces	256	484
closed directly	221	450
closed through all their continuations	35	34
continuations required per piece	16	22
satisfiable pieces found	0	0
pieces left undecided	0	0

The aggregator builds the expected set of pieces from the manifest rather than from the results, treats a missing piece and an undecided one as failures, and distinguishes three outcomes rather than two: only an explicit unsatisfiable answer closes a piece, a satisfiable one anywhere would destroy the claim, and a killed or timed-out run counts as *no information* — it neither closes nor refutes. Eleven such uninformative lines occur in the $n = 6$ record, from runs interrupted when work was moved between machines; each of the pieces concerned is closed by an explicit unsatisfiable answer from another run.

The lower bounds are witnesses, checked by exhaustive integer arithmetic over all $\binom{|S|}{3}$ triples by a program sharing no code with the search: 40 points at $n = 5$ (9880 triples) and

64 points at $n = 6$ (41664 triples). For $n = 6$ two witnesses were obtained independently, by a SAT solver and by a constraint solver in decision form; they turned out to be the same set, which the stabiliser explains — it has order 24, so the configuration has only two distinct images under the 48 symmetries of the cube.

As a further check that is not part of the proof, a constraint solver was given the $n = 6$, $M = 65$ question in one piece for an hour and returned no answer. That is not evidence of impossibility — it is the solver saying it does not know — but it found no counterexample either, while at $M = 64$ it produces a solution in seconds.

3.3 Structure of the optima, and an anomaly at $n = 5$

Writing the layer profile of a set as the number of its points in each of the n layers perpendicular to an axis, every optimum we have found has the *same* profile along all three axes:

n	$a(n)$	layer profile (identical for all three axes)
2	8	4, 4
3	16	6, 4, 6
4	28	8, 6, 6, 8
5	40	8, 8, 8, 8, 8
6	64	12, 10, 10, 10, 10, 12

A layer is an $n \times n$ grid with no three collinear, so it holds at most $2n$ points. For $n = 2, 3, 4, 6$ the two outer layers attain that planar maximum and the inner ones hold $2n - 2$, giving $2 \cdot 2n + (n - 2)(2n - 2) = 2n^2 - 2n + 4$ — which reproduces 8, 16, 28 and 64 exactly.

At $n = 5$ the pattern fails, and it fails for a structural reason rather than by accident. Prescribing the profile as a hard constraint and deciding feasibility shows that

profile at $n = 5$	verdict
10, 8, 8, 8, 10 (total 44, the value the pattern predicts)	infeasible
10, 9, 8, 9, 10; 10, 8, 9, 8, 10; 10, 10, 10, 10, 10; 9, 9, 9, 9, 9	infeasible
10, 8, 8, 8, 8 (total 42) and 10, 7, 8, 7, 10 (total 42)	infeasible
10, 6, 8, 6, 10 (total 40)	infeasible
8, 8, 8, 8, 8 (total 40)	feasible
$n = 6$: 12, 10, 10, 10, 10, 12 (total 64)	feasible
$n = 7$: 14, 12, 12, 12, 12, 12, 14 (total 88)	infeasible

A caveat on how these verdicts are to be read. Prescribing the profile along all three axes at once is a restriction, so an infeasibility obtained that way would exclude only symmetric configurations, not that number of points. Every infeasibility above was therefore re-derived with the profile prescribed along a *single* axis, which excludes every configuration with that layer distribution. The satisfiable rows need no such care: a solution is a solution. The fourth line is the sharpest: at $n = 5$ two outer layers cannot both attain the planar maximum $2n$ at *any* total, so the optimum is forced onto a uniform profile. The same computation at $n = 6$ finds the analogous profile realisable, and at $n = 7$ it is again impossible. The pattern $[2n, (2n - 2)^{n-2}, 2n]$ is thus realisable at $n = 2, 3, 4, 6$ and impossible at $n = 5, 7$ — the obstruction is not an artefact of one case, and it is not an obstruction to the idea of maximal outer layers as such.

The obstruction can be priced exactly. A layer is a planar no-three-in-line configuration and so carries at most $2n$ points; requiring *both* outer layers to attain that planar maximum, and maximising the total, gives

n	both outer layers	maximum total	$a(n)$
4	8	28	28 — costs nothing
5	10	39	40 — costs exactly one point
6	12	64	64 — costs nothing

So the anomaly is not that a fitted formula occasionally fails. It is that two planar-optimal configurations placed in the outer planes are compatible with the spatial optimum at $n = 4$ and $n = 6$ and incompatible at $n = 5$, and incompatible by a single point.

We do not have a proof of why. A count of the planar configurations attaining $2n$ points suggests rigidity: the 4×4 grid admits 11 such configurations in 4 classes under the square’s symmetries, the 5×5 grid admits 32 in only 5 classes, and the 6×6 grid admits 50 in 11 classes. We offer this as an observation accompanied by its data, not as an argument. One natural mechanism can be ruled out: the midpoint of a segment joining the two outer layers is a lattice point when n is odd, and must stay empty, but an exact count shows the middle layer still retains up to 16 free cells — above its own planar ceiling of 10 — so that constraint does not bind.

3.4 Certified lower bounds for $n = 7, \dots, 11$

n	$a(n) \geq$	$/n^2$	source of the witness
7	73	1.49	fourteen inequivalent configurations known; the first was invariant under every permutation of the axes (order 6)
8	94	1.47	two inequivalent configurations, each invariant under a subgroup of order 4 (an earlier version had 93, order 6)
9	116	1.43	two inequivalent configurations: stabilisers of order 4 and of order 12; the latter is optimal within its class
10	138	1.38	full stabiliser of order 12; optimal within the order-12 class
11	164	1.36	stabiliser of order 12; the bound inside the class is not closed

The bounds for $n = 8, \dots, 11$ (2–3 September 2026) come from a sweep over all 33 conjugacy classes of subgroups H of the symmetry group of the cube (order 48): in each class the maximum of $|S|$ over H -invariant admissible sets is computed by exact optimisation (CP-SAT over the H -orbits of cells, with “at most two chosen cells on every grid line”) under a time limit of 25–40 minutes per class. Where the solver closes the bound the value is the exact maximum of the class (this happened for the order-12 class at $n = 9$ and $n = 10$); elsewhere it is a lower bound inside the class. Every configuration was verified by two independent programs over all $\binom{|S|}{3}$ triples, its stabiliser computed under the 48 symmetries, and its local optimality certified: no point can be replaced (all are rigid, $\min \kappa \geq 2$ in the sense of the disjoint-killers lemma of the companion repository `lemma-atelier`), and no exchange removing at most four points (73, 94, 116) or three points (138, 164) improves it. The configurations, the per-class results and the verifier are archived separately [20]; the six witnesses for $n = 8, \dots, 11$ are among the ancillary files of this article.

The rest of this subsection is the earlier account of $n = 7, 8$ by randomised search, kept as data. Each configuration is re-verified by testing all $\binom{|S|}{3}$ triples with integer cross products. These are lower bounds only; we make no claim about how far they are from the truth, and we expect them to be weak, since a randomised search is a poor instrument at this size. The

$n = 7$ bound illustrates the point: it stood at 71 from randomised search and rose to 73 within the hour once the search was restricted to configurations invariant under a cyclic permutation of the axes. Restricting to a subgroup collapses the cells into orbits and shrinks the search by the order of the group; it is legitimate for lower bounds only, since a configuration found is a configuration, while failure to find one among the symmetric ones says nothing about the rest. The size of the group has an optimum in the middle rather than at either end, and the optimum is a balance: the group must be large enough to make the problem solvable and small enough that good configurations still fall inside the symmetric stratum. At $n = 8$:

order of the group	best found
2	82, 84
3	88
4	88, 90
6	92, 92, 93
8	80, 88
12	88
24	88 (proved optimal within its stratum)
48	80 (proved optimal within its stratum)

(The full sweep above, 40 minutes per class, later gave 94 in two classes of order 4, so the order-6 value 93 of this table was not the maximum of its own neighbourhood of classes either.) At $|G| = 48$ only 20 orbits remain, the solver settles the stratum in seconds, and the answer is 80, because genuinely good configurations do not have that much symmetry. Note also that the order alone does not decide it: two different groups of order 8 gave 80 and 88. We record this as data. An earlier draft of this part stated instead that a larger group simply gives a better result — a pattern read off the first four rows, which the remaining four contradict. (An earlier draft also quoted 72 and 90 here. Those numbers came from a run whose witness was never saved, and when we came to attach witnesses to them the journals we could reproduce gave 71 and 89. We report the values we can exhibit.) Two indications. First, the layer structure of the smaller optima suggests $2n^2 - 2n + 4 = 88$ at $n = 7$ — and although we have shown that the corresponding *profile* is impossible, that leaves the value itself untouched. Second, at $n = 6$ the same randomised search reached 64, which the certified computation then confirmed to be optimal, so the instrument is not hopeless either; we simply do not know which case $n = 7$ resembles.

For the record we also measured what the abandoned route would have cost, though the measurement deserves a caveat. Knuth’s random-probe estimator, calibrated against the exact tree at $n = 4$, put an unaided exhaustive proof at $n = 6$ at some $4.6 \cdot 10^{22}$ nodes, and that figure is what sent us to a decision procedure. We later found the plain estimator unreliable on trees of this family — on a sibling problem the mean of twenty thousand probes was 89% a single probe, with the median five orders of magnitude below the mean — and replaced it by a stratified version that agrees with the exact tree at $n = 4$ to within 0.1%. The $4.6 \cdot 10^{22}$ was produced by the plain estimator and has not been recomputed with the stratified one, so it should be read as an order of magnitude and nothing finer. It says nothing about the difficulty of the problem in any case, only about the difficulty of one way of attacking it.

3.5 An elementary lower bound for every prime

Proposition 3.1. *For every prime p , $a(p) \geq p^2$.*

Proof. Let $Q(x, y) = x^2 - dy^2$ with d a quadratic non-residue modulo p , so that $Q(a, b) \equiv 0$ only for $a \equiv b \equiv 0$ (an anisotropic binary form; one exists for every p). Put $S = \{(x, y, Q(x, y) \bmod p) : 0 \leq x, y < p\}$, of size p^2 . Suppose three of its points are collinear. Their projections to the

(x, y) -plane are distinct (the third coordinate is a function of the first two), hence lie on a line with primitive integer direction (a, b) , and the three points are $(x_0 + t_i a, y_0 + t_i b, \cdot)$ with distinct integers t_i , $|t_i| < p$. Along this line the third coordinate satisfies $z \equiv Q(x_0 + ta, y_0 + tb) = Q(a, b) t^2 + \ell(t) \pmod{p}$ with ℓ affine, whereas collinearity in $\mathbb{R}^3$ forces z to be an affine function of t . A quadratic congruence agreeing with an affine one at three points distinct modulo p has vanishing leading coefficient, so $Q(a, b) \equiv 0$ and therefore $(a, b) = (0, 0)$, a contradiction. $\square$

Neither $x^2 + y^2$ (anisotropic iff $p \equiv 3 \pmod{4}$) nor $x^2 + xy + y^2$ (iff $p \equiv 2 \pmod{3}$) works for all p ; the choice $x^2 - dy^2$ does. Proposition 3.1 is thereby the completion, to every odd prime, of Lemma 4 of Pór and Wood [13], whose construction is the case $Q = x^2 + y^2$ and works precisely for $p \equiv 3 \pmod{4}$; we verified their dichotomy directly (6 and 286 collinear triples at $p = 5$ and $p = 13$, none at $p = 7, 11$). For $p = 2$ the statement is trivial: no line meets $\{0, 1\}^3$ in three points, so $a(2) = 8 > 4 = p^2$. The bound of Proposition 3.1 is not sharp: $a(5) \geq 40 > 25$ and $a(7) \geq 73 > 49$. As a guard against a slip in transcribing the construction rather than in the proof, we also built S explicitly for $p = 3, 5, 7, 11, 13$ and tested *every* one of its triples by integer cross product — 790 244 triples at $p = 13$ — finding none collinear.

3.6 What is not established in Part I

The following are the limits of what is established in this part, stated so that a reader need not infer them.

- At $n = 5$ the upper bound no longer rests on one solver: the same 256-piece split was closed independently by Glucose [16], of the MiniSat lineage rather than kissat's, with every piece unsatisfiable and none satisfiable, the last taking 2785 seconds and 14.4 million conflicts. Two solvers of unrelated descent, two independently written aggregators, two independent semantic checks and two independently verified witnesses now stand behind $a(5) = 40$. Worth recording separately: the *monolithic* instance defeated Glucose for 10,000 seconds while its pieces fell in minutes, so the case split is a property of the method rather than of one solver. **At $n = 6$ this has not been done** — there are 484 pieces and they are heavier — and the reader should not carry $n = 5$ over to it.
- The upper bound for $n = 6$ rests on one SAT solver. The instances that a single call decides carry DRAT certificates verified by `drat-trim`, which removes the dependence on the solver; the instances that had to be split do not carry stored certificates, because the certificates run to tens of gigabytes. They are regenerated deterministically on demand, and the case split, its completeness and its aggregation are each verified separately.
- We have no proof of the upper bounds independent of the propositional encoding. Our own exhaustive enumerator does not reach $n = 5$: it closed 24 of 4096 prefixes at 15–33 billion nodes each, with the full traversal on the order of 10^{14} nodes. It corroborates the lower bound only.
- The verification of the symmetry pruning shows that our *implementation* discards nothing; it is not a proof of the underlying lex-leader argument, which is standard but which we did not reprove.
- The stratum sweep of Section 3.4 proves nothing about $a(n)$: a class maximum is an upper bound for that class only, and the classes are enriched in maxima but need not contain them — at $n = 6$ every class with a non-trivial symmetry stays below 64 except those attained by the certified optimum itself.
- The layer-structure anomaly at $n = 5$ and $n = 7$ is a computation, not an explanation. A count of the planar configurations attaining $2n$ points suggests rigidity (11 configurations in 4 classes for 4×4 ; 32 in only 5 classes for 5×5 ; 50 in 11 classes for 6×6), and one

natural mechanism — the lattice midpoint between two outer layers — is ruled out by an exact count. Beyond that we do not know.

4 Part I': the cube, no four coplanar

4.1 Prior art, and a claim of earlier versions corrected

Earlier versions of this part claimed that published data for $b(n)$ stop at $b(8) = 20$. That claim was false, and it is corrected here rather than erased: two sources predate this work, and both sit in plain sight on the OEIS page this part cites.

The 2017 comment. The entry A280537 carries, since January 2017: “Currently (January 2017) known lower bounds for the next terms are $a(9) \geq 23$, $a(10) \geq 26$, $a(11) \geq 28$, $a(12) \geq 30$, $a(13) \geq 32$, $a(14) \geq 35$, $a(15) \geq 36$, $a(16) \geq 38$, $a(17) \geq 42$ ” [18] (the entry’s a is our b). The same entry states that terms up to $b(6)$ were found by exhaustive search and that “ $a(7)$ and $a(8)$ are based on extensive numerical evidence” — which also corrects an error repeated in Section 4.5 by earlier versions.

The 2016 contest. Al Zimmermann’s programming contest *Non-Coplanar Points* (March–June 2016) posed exactly this problem for the first twenty-five primes [19]. The final report lists, for each size, the best score achieved outside the first 48 hours; the score is the square of the number of points, so the per-size bests are recoverable exactly:

n	best	n	best	n	best	n	best	n	best
2	5	13	32	31	71	53	110	73	145
3	8	17	42	37	81	59	121	79	157
5	13	19	46	41	89	61	125	83	164
7	18	23	54	43	92	67	137	89	173
11	28	29	66	47	100	71	142	97	188

(The full per-size decode from the final report, verified against the exact values 5, 8, 13, 18 at $n = 2, 3, 5, 7$. Where this work overlaps: ties at 11, 13; below by 4, 3, 4, 7 at 17, 19, 23, 29.)

The contest reached ratio $2.67n$ at $n = 3$ falling to $1.94n$ at $n = 97$; whether that decline reflects the problem or the budget is precisely the question Section 4.6 argues search cannot settle — but the numbers themselves are prior art and stand.

The comparison for every row of this work.

n	this work	prior (source)	verdict
9	23	23 (2017)	tie, witness now public
10	26	26 (2017)	tie, witness now public
11	28	28 (2017; 2016)	tie, witness now public
12	31	30 (2017)	improved by 1
13	32	32 (2017; 2016)	tie, witness now public
14	34	35 (2017)	<i>below prior art</i>
15	35	36 (2017)	<i>below prior art</i>
16	38	38 (2017)	tie, witness now public
17	38	42 (2017; 2016)	<i>below prior art by 4</i>
18	41	42 (monotone from $b(17)$)	<i>below the monotone consequence</i>
19	43	46 (2016)	<i>below prior art</i>
20	45	46 (monotone from $b(19)$)	<i>below the monotone consequence</i>
21	47	46 (monotone from $b(19)$)	improved by 1
22	49	46 (monotone from $b(19)$)	improved by 3
23	50	54 (2016)	<i>below prior art</i>
24	52	54 (monotone from $b(23)$)	<i>below the monotone consequence</i>
25	53	54 (monotone from $b(23)$)	<i>below the monotone consequence</i>
27	56	54 (monotone from $b(23)$)	improved by 2
29	59	66 (2016)	<i>below prior art by 7</i>

Values marked “monotone” are the closure of the published bounds under $b(n+1) \geq b(n)$ (an n -cube configuration is also an $(n+1)$ -cube configuration). An earlier version of this table compared against the published rows only and marked the composite sizes “apparently first published”; the monotone comparison is due to Hugo Pfoertner. The 2017 and 2016 numbers come without accessible configurations — the contest server verified scores, but the point sets themselves were not published; every number of ours comes with a public configuration, verified twice by programs sharing no code. *Certified public witnesses* against *score-only records* is the honest division of value, and it is why the rows below prior art stay in the table: they carry the only accessible configurations at their sizes that we know of.

One more thing the comparison measures. Our search above $n = 16$ was confined to the cyclically invariant subspace, and the gap to the 2016 records is 4, 3, 4, 7 exactly on the primes where they exist: an external measurement of what that confinement costs at growing n , consistent with Section 4.3 — enrichment is not containment, and here its price has a number.

How this was missed is worth recording, because the mechanism is general: the OEIS DATA line does stop at $b(8) = 20$, and we read the line and not the page. The comment with nine further bounds sat directly under it, and the contest is linked from the same entry. An external review pointed both out.

4.2 The bounds

n	lower bound	$3n$	gap	quadruples checked
9	23	27	4	8 855
10	26	30	4	14 950
11	28	33	5	20 475
12	31	36	5	31 465
13	32	39	7	35 960
14	34	42	8	46 376
15	35	45	10	52 360
16	38	48	10	73 815
17	38	51	13	73 815
18	41	54	13	101 270
19	43	57	14	123 410
20	45	60	15	148 995
21	47	63	16	178 365
22	49	66	17	211 876
23	50	69	19	230 300
24	52	72	20	270 725
25	53	75	22	292 825
26	53	78	25	292 825
27	56	81	25	367 290
28	56	84	28	367 290
29	59	87	28	455 126

Twenty-one rows, but only nineteen configurations. The rows for $n = 26$ and $n = 28$ carry no configuration of their own: they follow from $n = 25$ and $n = 27$ by monotonicity alone, and are printed only to keep the table contiguous. The row for $n = 17$ does have its own configuration, but no longer carries its own information, since $b(17) \geq b(16) \geq 38$ follows by monotonicity. This has happened three times as the neighbouring row was improved, and it is the reason to read the table as a set of bounds rather than of values — a row is worth something only while it exceeds its left neighbour, and against prior art and its monotone closure four rows do (Section 4.1).

The configurations themselves are in the repository (Section 8). The gap to Proposition 2.1 grows with n ; whether this reflects the problem or the reach of the search we cannot say, and Section 4.6 explains why we believe the question is not decidable by search.

4.3 The cyclically invariant subspace

All configurations above were found by searching the subspace invariant under a cyclic permutation of coordinates. That subspace is enormously enriched in maxima, and enriched more as n grows. In the two-dimensional analogue, where we hold complete material — every maximum configuration for $n \leq 11$, reproduced against A000755 — the proportion of maxima carrying a symmetry of the square, divided by the same proportion in a null model matched on row and column counts, runs

$$\times 8, \times 56, \times 125, \times 769, \times 3729, \times 73\,000 \quad (n = 6, \dots, 11),$$

the last figure resting on seven events in four million and therefore stated to one significant figure only. The raw proportion *falls* across the same range, from 36% to 13%; the ratio to the null model rises.

Enrichment is not containment. At $n = 6$ in three dimensions the cyclically invariant subspace contains no maximum at all: its exhausted maximum is 15 while $b(6) = 16$. Exhaustive

traversal of that subspace gives exactly 23 at $n = 9$ and exactly 26 at $n = 10$, each exhaustion carried out by two implementations written without sight of each other's code (Section 8); these two statements bound the subspace and say nothing about $b(9)$ and $b(10)$ beyond the witnesses.

One consequence deserves emphasis. A sample drawn from a region enriched by three or four orders of magnitude cannot support statements about the structure of solutions in general. Structural claims in this part are made only where complete material is available.

4.4 Symmetry strata

A second search route — the stratum sweep of Section 3.4, with the plane constraints added — gives three structural facts and changes no bound in the table.

Route. Take every conjugacy class of subgroups H of the symmetry group of the cube (order 48; 33 classes), and in each class maximise $|S|$ over H -invariant sets by exact optimisation: a CP-SAT model over the H -orbits of cells with the constraints “at most two points on every grid line” and “at most three on every lattice plane carrying at least four cells”, the plane list computed exactly (12 453, 73 056, 269 895, 965 826, 2 688 585, 7 241 856 planes for $n = 5, \dots, 10$). A lazy version of the model, which adds planes only when a solution violates them, does not converge (at $n = 6$ it stalls at 12 after two hundred rounds) — the full list is necessary.

Four kinds of symmetry are incompatible with the problem outright. If H contains the central inversion or a rotatory reflection of order 6, an H -invariant admissible set has at most 3 points (two antipodal pairs are coplanar); if H contains a reflection, at most 5 (the segments $x\text{--}\sigma x$ are parallel, so any two such pairs are coplanar, and the mirror carries at most three points); if H contains a rotation of order 4, the set lies on the axis. Only nine of the 33 classes remain: the trivial group, two classes of half-turns, the 3-fold rotation, two Klein four-groups, the cyclic group of a rotatory reflection of order 4, the dihedral group of order 6 and the tetrahedral rotation group. (The elementary proofs, with machine checks and a Lean formalisation of the inversion case, are in the companion repository `lemma-atelier`, lemma 003.)

Calibration. Within the strata the known values are reached at $n = 5, 7, 8$ (13, 18, 20) by the classes containing a 3-fold rotation or a half-turn; at $n = 6$ the best symmetric value is $15 = b(6) - 1$, in agreement with Section 4.3. At $n = 9, 10$ the 3-fold stratum reproduces 23 and 26 — the subspace maxima already exhausted by the first route — with a one-hour budget, and nothing above them. The trivial stratum (no symmetry imposed) reached 17 at $n = 7$ and 19 at $n = 8$ within 900 seconds and proves nothing.

Structural facts, every configuration checked by the two verifiers of Section 8 and its class computed under the 48 symmetries: (i) at $n = 7$ there are at least three pairwise inequivalent 18-point configurations, with stabilisers of orders 1, 2 and 3; at $n = 8$ three inequivalent 20-point configurations (orders 1, 2, 3); at $n = 9$ three inequivalent 23-point configurations (all of order 3). The order-2 classes lie outside the cyclically invariant subspace of Section 4.3. (ii) All nine are rigid — no point can be replaced by another — and no two are connected by an exchange of at most two points (at most three points for $n = 7$); under the best alignment the classes differ in at least 80% of their points (16 of 20 for one pair at $n = 8$, more for the others). (iii) Every stratum at $n = 5, \dots, 8$ was solved to optimality or to the time limit; the per-stratum table and the configurations are in the repository, and seven of the configurations (the classes of orders 2 and 3 at $n = 7, 8$, two of order 3 at $n = 9$, and a 28-point configuration of order 3 at $n = 11$) are among the ancillary files of this article.

What this does not establish: the strata give lower bounds inside symmetry classes only, and the strata at $n \geq 11$ (best known 28 and 31) have not been exhausted.

4.5 What is not established in Part I'

No configuration proves an upper bound. A witness certifies “at least” and can never certify “at most”. The values $b(5) = 13$ and $b(6) = 16$ rest on exhausted search trees; $b(7) = 18$ and $b(8) = 20$ rest, per the OEIS entry, on “extensive numerical evidence” only — an earlier version of this part wrongly promoted them to exhausted; above $n = 8$ we have Proposition 2.1 and nothing else. In particular the numbers in the table are not claimed to be maxima, and we expect several of them to be improved. A first-moment estimate of ours that these witnesses refuted is recorded in Section 7.1.

4.6 Effort, and why the growth rate is not measurable by search

The ratio of certified bound to n decreases over the range of the table. We do not read this as a property of the problem. Giving each seed an equal time budget on one core (eight seeds, fifteen seconds of uniform random greedy restarts each, the cell drawn uniformly among the alive) and recording how many seeds reach the OEIS value, we measure $8/8$ at $n = 5$ and $0/8$ at $n = 6, 7, 8$ — at $n = 6$ none of roughly 160 000 restarts per seed reaches 16. *Correction:* versions of this part up to 2.9 quoted $8/8, 3/8, 2/8, 1/8$ here; those four fractions were measured on the *planar* no-three-in-line problem — they are the ones reported in Section 5.7 — and migrated into the cube account without a cube re-measurement. The corrected cliff is steeper, which strengthens the point but does not excuse the migration. Equal time is therefore not equal effort, and a curve obtained under a fixed budget records the budget. Deciding whether $b(n)/n$ tends to 3 or to something smaller would need a budget growing exponentially in n .

A second contamination is worth naming. In a search whose target doubles as its stopping condition, asking for K and receiving K is not evidence about K . Several values in an earlier draft of this table were produced that way, agreed with a prediction, and were withdrawn when the target was removed and the same program returned more.

5 Part II: the plane — the Guy–Kelly heuristic against exact counts

5.1 The heuristic, and which constant is which

The number $T(n)$ of collinear triples in the $n \times n$ grid has a closed form,

$$T(n) = \sum_{\gcd(a,b)=1} \sum_{s \geq 2} (s-1)(n-s|a|)^+(n-s|b|)^+,$$

which we verified against brute force on eighteen values. The heuristic estimates the number of admissible m -subsets by

$$\binom{N}{m} \exp\left(-T(n) \binom{m}{3} / \binom{N}{3}\right), \quad N = n^2, \quad (2)$$

and locates the threshold where this drops below one. Guy and Kelly’s original constant was $(2\pi^2/3)^{1/3} = 1.8739$; Gabor Ellmann found an error in the argument in March 2004, and the corrected value, recorded by Guy in OEIS A000769 that October and documented by Voutier [7], is

$$c = \pi/\sqrt{3} = 1.813799.$$

Our own derivation of the threshold reproduces the corrected value, not the retracted one. As a check of the machinery we located the crossing exactly: the threshold from (2) first falls below $2n$ at $n = 493$ (at $n = 492$ it equals $984 = 2n$; at $n = 493$ it is $985 < 986$), which agrees to the integer with the value reported by Prellberg [5] and quoted in the recent survey [4]: “the

heuristic probabilistic argument only applies for $n \geq 493$ ". Our computation is independent of his and reproduces the same integer.

5.2 The constant in closed form, and the shape of the slip

The corrected value can be obtained in closed form, and doing so shows exactly what the retracted one is. The input is the asymptotics of $T(n)$. Measuring $T(n)/n^4$ at $n = 2000, \dots, 64000$, its increment per doubling of n is constant to six figures over five doublings — 0.210690, 0.210692, 0.210693, 0.210692, 0.210691 — against

$$(3/\pi^2) \ln 2 = 0.210691,$$

so that

$$T(n) = \frac{3}{\pi^2} n^4 (\ln n - 0.8373) + o(n^4 \ln n). \quad (3)$$

Put $m = \alpha n$. Then $\ln \binom{n^2}{m} = \alpha n (\ln n + 1 - \ln \alpha) + O(n)$, while by (3) the exponent of (2) is $T(n)(m/n^2)^3 = \frac{3}{\pi^2} \alpha^3 n (\ln n - 0.8373) + O(n)$. Both sides are $\Theta(n \ln n)$; balancing the coefficients of $n \ln n$ gives

$$\alpha = \frac{3}{\pi^2} \alpha^3, \quad \text{i.e.} \quad \alpha^2 = \pi^2/3, \quad \alpha = \pi/\sqrt{3}.$$

The balance is quadratic. Guy's $(2\pi^2/3)^{1/3}$ is the root of $\alpha^3 = 2\pi^2/3$: a cubic where a quadratic belongs. That is the whole of the slip, and it is visible only once (3) is in hand, which is presumably why it stood for thirty-six years.

Numerically, the integer threshold $m^*(n)/n$ approaches the limit only logarithmically, so a single large n proves nothing. Two implementations built independently — one summing over displacement vectors, one reducing the same sum to $O(n \log n)$ by the Dirichlet identity $\gcd = \sum_{d|\gcd} \varphi(d)$ — agree to 10^{-5} from $n = 16000$ on:

n	first	second
16000	1.93101	1.93100
32000	1.92307	1.92306
64000	1.91615	1.91614

Fitting $m^*/n = L + A/\ln n$ gives $L = 1.8126$ and 1.8118 ; adding $B/(\ln n)^2$ gives 1.8154 and 1.8142 ; a third correction term destabilises on seven points and is not used. The two orders bracket from below and from above in *both* implementations, giving

$$L = 1.8138 \pm 0.002,$$

which contains $\pi/\sqrt{3} = 1.813799$ and excludes $(2\pi^2/3)^{1/3} = 1.873856$ by thirty times the uncertainty. We state two decimals of confidence and no more: three decimals are tempting here and are not supported once the third fitting term is seen to wander.

5.3 Testing the count, not the threshold

A threshold inherits the whole error of the estimate without exhibiting its size. The count does not. Exact counts of $2n$ -point configurations are A000755, published to $n = 19$; the value at $n = 20$ is on A. Flammenkamp's table. We obtain both independently by summing orbit sizes $8/|\text{stab}|$ over the D_4 -classes of his solution database, which reproduces $s(2), \dots, s(19)$ exactly and gives $s(20) = 941580$ (since entered in A000755). Against (2):

n	exact	estimate	$\log_{10}(\text{exact}/\text{estimate})$
6	50	$10^{4.12}$	-2.421
10	1135	$10^{7.11}$	-4.055
14	10568	$10^{9.63}$	-5.604
18	152210	$10^{11.78}$	-6.597
20	941580	$10^{12.87}$	-6.896

The error reaches seven orders of magnitude. Its increment per step, however, decays by a factor of four across the range: $-0.86, \dots, -0.25, -0.13$. That distinction — between an error whose increment is constant and one whose increment decays — is what the rest of this part turns on.

5.4 Refining the heuristic makes it worse

Triples on one line are strongly dependent. The natural repair is to treat *lines* as independent and use the exact hypergeometric probability that a line of k grid points carries at most two chosen points. It is worse, and by a clean factor:

	triples independent	lines independent
$n = 6, m = 12$	-2.421	-4.107
$n = 7, m = 14$	-2.639	-5.285

The mechanism is that (2) carries *two* errors of opposite sign. Independence *between* lines overestimates, and the measured direction fixes the sign of the correlation: at the ceiling $m = 2n$ the two-per-row budget is rigid, so a set that has avoided one line finds the next *harder* to avoid — the events are negatively correlated, and a product of marginals overshoots. (An earlier draft asserted the opposite sign while drawing the same conclusion; the direction stated here is the one the numbers force.) This decomposition is measured at the ceiling only: away from it the net error changes sign (Section 5.6), and we do not claim the two-error account transfers there. Counting the $\binom{k}{3}$ overlapping triples on a line as independent events overestimates the probability of a violation, hence underestimates the survival probability. Replacing the second by an exact computation removes the compensation and leaves the first bare. At $n = 7$ the cancellation is worth $10^{2.65}$.

The practical corollary is uncomfortable: any refinement that fixes one of the two errors will make the heuristic worse. Only fixing both helps, and the first requires the correlations between lines, which is precisely what nobody knows how to compute. This may be why the 1968 form has not been superseded in fifty-eight years.

The same mechanism predicts where the heuristic should fail hardest, and we can test that prediction on our own data. In a symmetric class the points are rigidly coupled, so the first error grows while the second does not, and the cancellation should break. The coupling was measured directly in Section 4.3: among all maximum configurations (complete enumeration, $n \leq 11$), those carrying a symmetry of the square are over-represented relative to a null model matched on row and column counts by factors of 8 to 3729 at $n = 6, \dots, 10$. Where the enrichment is three orders of magnitude, independence between constraints is not a small approximation, and a heuristic built on it should not be trusted — as reportedly it is not, for symmetric classes.

5.5 Which error is which class

Over sixteen steps the line-independent version has an essentially constant increment, -1.6 per step, while the triple-independent version decays fourfold. A constant increment means an error growing exponentially in n , i.e. a wrong exponent; a decaying increment means a wrong

multiplier only. So the refinement is not merely further from the truth on a given n : it is wrong in a different and worse way.

We add what this does *not* yield. It is tempting to calibrate: subtract the measured error from (2) and re-solve for the n at which $2n$ -point configurations should cease to exist. We carried that out and then withdrew it. Fitting the error on $n = 12, \dots, 20$ against $\ln n$, $\sqrt{n}$, $n^{3/4}$, $n/\ln n$ and n gives root-mean-square residuals of 0.133, 0.166, 0.186, 0.194, 0.207 — nine points do not separate five one-parameter families — and calibrated crossings ranging over a factor of three. Worse, the spread is a property of the family we chose to fit, not of the data: $n/(\ln n)^2$ and $n^{0.9}$ are equally admissible, equally $o(n \ln n)$, and give further values. The honest statement is that the finite- n crossing is not determined by the data at all, and we give no number for it, because a number given here would be quoted later as a measurement.

What the data do determine is the direction that matters. Every form compatible with the nine points is $o(n \ln n)$, and the terms of the heuristic are $\Theta(n \ln n)$. A correction of lower order does not move the balance of Section 5.2, so the discrepancy — seven orders of magnitude though it is at $n = 20$ — leaves $\pi/\sqrt{3}$ where it is.

5.6 At fixed ratio the multiplier is not bounded, and the growth form cannot be chosen

An earlier version of this part reported that at fixed ratio $r = m/2n = 0.9$ the error of (2) stops growing with n , and read that as thirteen standard deviations of evidence that the exponential rate of the heuristic is correct. We withdraw the measurement, the inference, and the significance, and we record why each failed, since the failures are of general kinds.

The measurement: an instrument that did not report its own reach

The estimator's *hit rate* — the fraction of random descents that reach depth m at all — was not recorded. It should have been. At $r = 0.9$ the two points carrying the claim were obtained at hit rates of 0.08% and 0.003%: one hundred and sixty successful descents at $n = 15$, and *five* at $n = 20$, per two hundred thousand attempts. A quoted uncertainty of ± 0.144 in $\log_{10}$ is not attainable from five hits by any number of seeds. Re-running that cell independently returns $1.15 \cdot 10^{20}$ against the earlier $2.03 \cdot 10^{19}$. Neither number carries information. We note that the values were never shown to be *wrong*; they were unsupported, which is a different and more common defect.

Against this we measured the estimator itself where exact counts exist, and it is sound: at $n=10, m=18$ (hit rate 2.9%) ten seeds give +2.2%; at $n=8, m=15$, -0.5%; at $n=8, m=16$ — the ceiling, eighty-four hits — +1.8%. There is no bias to speak of. The defect is variance, not bias, and it is curable by descents rather than fatal.

The finding, once ratios with high hit rates are used

Writing $E(n, r) = \ln(\text{estimate of (2)}) - \ln(\text{true count})$ in natural logarithms, and holding r at values where $2rn$ is an integer for every n used — the ratio must not be rounded, since at $n = 20$ the error moves by 36 per unit of r , so rounding m shifts it by 0.45 —

r	E against n	hit rate	anchors
0.50	+0.09, +0.10, +0.14, +0.12, +0.08, -0.22, -0.69, -1.28, -1.96, -2.71, -3.51	100%	four exact
0.70	+0.58, +0.68, -0.09, -1.29, -2.80, -4.50	99.6%	one exact
0.80	+1.18, +1.71, +0.98, -0.33, -1.95, -4.58	$\geq 5\%$	one exact

($n = 5, 6, 7, 8, 10, 15, 20, 25, 30, 35, 40$ for the first row, continued on its second line; $n = 5, 10, 15, 20, 25, 30$ for the others.) At each of these ratios the error rises, turns, and then declines without levelling. The multiplier is not bounded: the heuristic understates the count by a factor growing at least exponentially in n , faster at larger r . The error is therefore a function of the *shape* of the question, not of n alone: at the single n where all three measurements exist ($n = 20$) the heuristic overestimates by a factor $8 \cdot 10^6$ at the hard ceiling $m = 2n$, by a factor of about 80 near the threshold ratio, and is off by less than 1.4 at $m = 1.6n$ (there an underestimate), where two independent implementations agree to 0.18 standard deviations.

The summit, however, moves right as r grows — near $n = 7$ at $r = 0.5$, near $n = 10$ at $r = 0.7$ and $r = 0.8$ — and at $r = 0.9$ it has not been reached at all. Five million descents at $n = 20$, $m = 36$ produce fifty-four hits and an estimate of $6.59 \cdot 10^{18} \pm 38\%$, giving $E = +4.42 \pm 0.32$ against the exact $E = +4.08$ at $n = 10$: still rising. (The earlier value $2.03 \cdot 10^{19}$ for this cell would have given $+3.30$; it was high by a factor of three, and rested on five hits.) So at the ratio that actually determines the constant — the threshold sits at $r = c/2 = 0.907$ — *we have never observed the decline*, only the ascent to a summit we cannot reach. Any statement about the behaviour of E near the threshold is therefore an extrapolation across a turning point that has not been located.

We record one further trap here, because we fell into it. On the first four (exact) points of the $r = 0.5$ row the error is flat, and we announced that the decline was a near-threshold phenomenon. It is not: those four points lie before the summit. Reading a pre-summit segment as a plateau is precisely the error being corrected in this subsection, and four exact values persuaded us of it more readily than eight estimated ones would have. Exactness of values is not length of series.

Why the growth form cannot be chosen

If $E = \Theta(n)$ the constant of Section 5.2 is untouched, since the terms balanced there are $\Theta(n \ln n)$. If $E = \Theta(n \ln n)$ it moves. The data do not decide, and cannot.

Fitting the full $r = 0.5$ row to $E = c + an + bn \ln n$ returns b at hundreds of standard deviations from zero — and a χ^2 of 240 on eight degrees of freedom, which rejects the model and voids the uncertainty with it. Fifteen further one-term variants using $\ln n$, $\sqrt{n}$ and $n/\ln n$ are also rejected, the best by a factor of eighty-four. The per-point error must be measured rather than assumed, and it does not scale: in the second implementation, eight independent seeds give 0.262% at $n = 30$, 0.404% at $n = 35$ and 0.979% at $n = 40$, so that the ratio of seed spread to batch spread runs 0.23, 0.42, 1.00 — no single measured value transfers to a neighbouring n . Our own repeats give 0.55% at $n = 30$ (two runs) and 2.98% at $n = 40$ (four seeds), against nominal figures of 1.3% and 1.8%: measured over nominal is 0.4 in one place and 1.7 in the other. Below, points with repeats carry the standard error of their mean and the rest carry the estimator's own figure, with no scaling between them.

We attempted to go further and failed, and the failure is worth recording because it is the error this part is otherwise about. The whole column was computed twice by implementations sharing no code, and comparing a single run of ours against the other implementation's value suggested a disagreement of 5.5% at $n = 40$ where the stated error was 0.98% — from which we concluded that the honest error bar is the cross-implementation one and cannot be had by seeds. A control refuted this. Four seeds of our own implementation at $n = 40$ give a spread of 3.0%, and their mean brings the disagreement with the other implementation down to 1.2%. The ratio is therefore 0.4, not 5.5: the cross-implementation difference is *smaller* than our own seed spread. Both halves of the original figure were wrong — the numerator was one outlying seed, and the denominator was the other implementation's error scale substituted for ours.

The a priori point survives: seeds cannot see what differs between implementations, since they vary only the random stream within one traversal order. But we have no measurement

supporting it, and we had presented it as measured. Below, each point uses the mean over the seeds we ran and their measured spread.

This matters beyond bookkeeping. With a single noisy seed at $n = 40$ and an error bar too small by a factor of three, every one of the four forms below is rejected — χ^2 of 27.6, 19.3, 24.8, 13.1 on two degrees of freedom — and we would have reported that no form fits. Averaging four seeds and using their measured spread turns that false certainty back into the correct verdict, which is that the forms cannot be told apart.

The rejection is confined to small n . Taken on tails, three-parameter forms are admissible from $n \geq 15$ onward: $\chi^2 = 0.7/3, 0.4/2, 0.3/1$ for $c + an + bn \ln n$ at $n \geq 15, 20, 25$. The asymptotic regime appears to begin near $n = 15$, and below it we were fitting asymptotic forms to pre-asymptotic data.

On that tail the fit is emphatic: $b = -0.0892 \pm 0.0021, -0.0872 \pm 0.0047, -0.0837 \pm 0.0109$ across three nested tails, while the purely linear $c + an$ is rejected at $\chi^2 = 1884/4, 350/3$ and $59/2$. The significance of b is 43, 19 and 8 standard deviations on the three tails — itself a warning, since a robust quantity does not lose three quarters of its significance when four points are dropped. Read alone this says $E = \Theta(n \ln n)$ and the constant moves. It does not say that. On the same tail,

form ($n \geq 20$)	χ^2 (2 d.o.f.)	implication
$c + an + bn \ln n$	0.44	constant moves
$c + an + d\sqrt{n}$	0.23	constant intact
$c + an + en/\ln n$	0.34	constant intact
$c + an + f \ln n$	0.34	constant intact

All four are admissible; three imply the opposite conclusion and fit better. An independent implementation, run without sight of these figures, obtains reduced χ^2 of 8.1, 7.0, 7.4, 7.5 for the same four forms — different in level, identical in verdict: mutually indistinguishable, oppositely answering. The forty-three standard deviations are a quantity computed *inside* a model, and such a quantity does not transfer to a choice *between* models. The test to apply is not whether a coefficient differs from zero, but whether a second admissible model exists in which it is absent. Here several do.

Nor is this curable by computing further. The two candidate bases are collinear to 0.9983 over $n \leq 40$, and the collinearity does not improve with range: 0.9986 to $n = 400$, 0.9993 to $n = 10^4$. What separates them is precision, not length, and the precision is already 0.26%.

We therefore withdraw the support that the earlier version of this subsection offered and do not replace it. The derivation of Section 5.2 stands, but it establishes what the first moment gives, not what the problem gives, and the distance between those two is what this subsection failed to measure.

5.7 Where the reach ends, and why

The limit is not an artefact of the estimator. A descent guided by importance sampling reaches only the depths a greedy search reaches, and the quality of greedy search degrades with n : we measured the fraction of random restarts attaining the known maximum as $8/8, 3/8, 2/8, 1/8$ at $n = 5, \dots, 8$, a geometric decay. At $n = 20$ greedy attains about 0.925 of the maximum, so the ratio 0.9 is within reach; by $n = 25$ it falls below 0.9 and the required depth is outside what such a descent visits at all. Every top-down greedy method shares this bound. To go past it one needs a descent guided by something that knows the target — which is to say, the upper bound that the problem does not have.

5.8 The objection this answers

Kaplan’s objection to the Guy–Kelly conjecture, as summarised in [8], is not that the constant is wrong but that the derivation is unsafe: it “suggests that certain random events would be largely independent of each other when they’re really not”, and “if we look more closely at how non-independent they actually are we might not reach the same conclusion”. No alternative constant is proposed.

The objection is exactly right about the derivation, and Sections 5.3 and 5.4 quantify how badly: the independence assumption is off by seven orders of magnitude at $n = 20$, and the crude form survives only because two of its errors cancel. We had believed Section 5.6 supplied the measurement the objection asks for. It does not, and we now believe no measurement of this kind can: the two possible answers correspond to bases that stay collinear at every range, and both admit an acceptable fit to the data we can obtain. To “might we not reach the same conclusion” the honest answer is that we cannot tell by counting, and that we have measured how far from telling we are.

6 Part II’: the direction spectrum of the same solutions

6.1 Definitions and data

For $v = (a, b)$ with $\gcd(a, b) = 1$ and $(a, b) \neq (0, 0)$, a *pair of direction* v in $S \subseteq [n]^2$ is an unordered pair $\{p, q\} \subseteq S$ with $q - p = kv$ for some $k \geq 1$ (so every pair has exactly one primitive direction up to sign). For a solution, every line carries at most two points, hence the number of pairs of direction v equals the number of lines of direction v carrying exactly two points. Define

$$c_v(n) = \frac{1}{n} \mathbb{E}[\#\{\text{pairs of direction } v\}],$$

the expectation over the uniform distribution on the solutions of size n in the sample at hand, averaged over the class $\{(a, b), (b, a), (a, -b), (b, -a)\}$ of v under the symmetries of the square. Since every row and every column of a solution carries exactly two points, $c_{(1,0)} = c_{(0,1)} = 1$ identically, and the total over all non-axial classes is fixed:

$$\sum_{v \text{ non-axial}} n c_v(n) \cdot |\text{class}(v)| = \binom{2n}{2} - 2n = 2n^2 - 3n. \quad (4)$$

Data. We use Achim Flammenkamp’s database of all known solutions (state of 31 August 2026). It is complete for $n \leq 20$ (for $n = 19$: 32 577 solutions; $n = 20$: 118 057) and contains only solutions with a non-trivial symmetry for $21 \leq n \leq 57$. The measurements were made by a second agent with code written independently of the model below; the values quoted as “mixed n ” average over all 391 812 solutions with weight one. Ensemble measurements on this database have been published since 1997 (Flammenkamp’s frequency maps of cell occupation [3], continued by Prellberg in 2026 for $n = 56, 57$); the present part continues that genre with a different observable.

class	mixed n	$n = 19-20$	$n = 29-31$	$n = 32-57$
(1, 1)	0.7306	0.7308	0.7327	0.7325
(1, 2)	0.5633	0.5638	0.5635	0.5548
(1, 3)	0.4545	0.4545	0.4586	0.4526
(2, 3)	0.4095	0.4097	0.4121	0.4070
(1, 4)	0.3744	0.3741	0.3807	0.3793
(1, 5)	0.3073	0.3074	0.3228	0.3261
(2, 5)	0.2846	0.2842	0.3023	0.3049
(3, 4)	0.3043	0.3039	0.3175	0.3186
(1, 6)	0.2618	0.2492	0.2687	0.2800
(1, 7)	0.2233	0.2071	0.2340	0.2461
(2, 7)	0.2081	0.1889	0.2208	0.2335
(3, 5)	0.2701	0.2609	0.2758	0.2819
(4, 5)	0.2390	0.2243	0.2486	0.2575
(1, 8)	0.1859	0.1692	0.1954	0.2160
(3, 7)	0.1961	0.1791	0.2069	0.2209

Short directions are constant in n to 1%; long directions grow with n (by 23% for (3, 7) and 28% for (1, 8) between $n \approx 20$ and $n \geq 32$); the sizes $21 \leq n \leq 28$ (54 604 solutions) are omitted from the table. A “constant of the database” for a long direction is therefore an average over a mixture of sizes, and the model must be compared with the data *at the same* n . Symmetry does not explain the growth: at $n = 19$ and 20 , where the database is complete, the spectrum of the solutions with a half-turn symmetry (592 and 675 of them) differs from that of all solutions by at most 4%, and for the long directions in the *downward* direction ($c_{(1,7)}$: 0.1986 against 0.2067 at $n = 19$).

The null model. Under the uniform distribution on all configurations with exactly two points in every row and column (the correct null model: a solution is such a configuration), the expected number of pairs of direction v is the number of cell pairs of direction v times the probability $2(2n-3)/(n(n-1)^2)$ that a given pair of cells in distinct rows and columns is occupied (the group $S_n \times S_n$ acts transitively on such pairs). At $n = 20$ this gives $c_{(1,1)}^{\text{null}} = 1.27$, $c_{(1,2)}^{\text{null}} = 0.75$, ..., so the solutions are depleted in the short directions (0.58, 0.75, 0.89, 0.94 for (1, 1), (1, 2), (1, 3), (2, 3)) and slightly enriched in the long ones (1.01–1.11). This is the pattern to be explained.

6.2 The line model

Fix a direction v . The cells of $[n]^2$ are partitioned into the lines of direction v (a cell without a neighbour in direction v is a line of length 1); let the lines have lengths $L_1, \dots, L_r$.

Theorem 6.1. *Let $2 \leq m \leq \sum_i \min(L_i, 2)$ and let S be uniformly distributed on the m -subsets of the cells that contain at most two cells of every line. Then the number of such subsets is $Z_m = [t^m] \prod_{i=1}^r (1 + L_i t + \binom{L_i}{2} t^2)$ and*

$$\mathbb{E}[\#\{i : |S \cap \ell_i| = 2\}] = \frac{1}{Z_m} \sum_{i=1}^r \binom{L_i}{2} [t^{m-2}] \prod_{j \neq i} (1 + L_j t + \binom{L_j}{2} t^2).$$

Proof. A subset is admissible for v exactly when it takes $x_i \in \{0, 1, 2\}$ cells on line i ; there are $\binom{L_i}{x_i}$ ways to choose them, independently across lines, so the admissible m -subsets are counted by the coefficient of t^m in the product. The expectation is $\sum_i P(x_i = 2)$, and the admissible m -subsets with $x_i = 2$ are counted by $\binom{L_i}{2}$ times the coefficient of t^{m-2} in the product over $j \neq i$. $\square$

This is the canonical ensemble of hard particles in r independent boxes of capacity two with degeneracies $1, L_i, \binom{L_i}{2}$ (equivalently, a uniform placement conditioned on “at most two per box”); the identity is elementary. We did not find an established name for it; the nearest umbrella term in the refereed literature is *restricted occupancy* (Freund 1956 [11]; the unweighted equal-box case is an exercise in Comtet), and the corresponding relaxation appears as “capacity-two constraints” in Prellberg’s linear-programming bound for a parity variant [6]. What is not elementary is that it describes the solutions of the no-three-in-line problem.

The model. Put $m = 2n$ and let E_v be the expectation of Theorem 6.1 for the lines of direction v . The model does not know that the pairs of a solution are counted by (4): summed over all non-axial primitive directions, $\sum_v E_v = 615.3$ at $n = 20$ against the true 740, a shortfall of 17%. We therefore impose (4) by one global factor, $\lambda_n = (2n^2 - 3n) / \sum_v E_v$ ($\lambda_{20} = 1.203$, $\lambda_{30} = 1.163$, $\lambda_{40} = 1.139$), and set $c_v^{\text{model}}(n) = \lambda_n E_v / n$, averaged over the class of v . The factor is not fitted to the measured spectrum — it is the same for every direction — but it is not derived either. Everything the model predicts is therefore a statement about *ratios* between directions.

6.3 Results

class	model 20	data 19–20	ratio	model 30	data 29–31	ratio
(1, 1)	0.7637	0.7308	1.045	0.7441	0.7327	1.016
(1, 2)	0.5629	0.5638	0.998	0.5541	0.5635	0.983
(1, 3)	0.4327	0.4545	0.952	0.4316	0.4586	0.941
(2, 3)	0.3766	0.4097	0.919	0.3792	0.4121	0.920
(1, 4)	0.3428	0.3741	0.916	0.3492	0.3807	0.917
(1, 5)	0.2781	0.3074	0.905	0.2890	0.3228	0.895
(2, 5)	0.2544	0.2842	0.895	0.2665	0.3023	0.882
(3, 4)	0.2737	0.3039	0.901	0.2839	0.3175	0.894
(1, 6)	0.2322	0.2492	0.932	0.2440	0.2687	0.908
(1, 7)	0.1931	0.2071	0.932	0.2105	0.2340	0.900
(2, 7)	0.1800	0.1889	0.953	0.1978	0.2208	0.896
(3, 5)	0.2309	0.2609	0.885	0.2443	0.2758	0.886
(4, 5)	0.2076	0.2243	0.926	0.2223	0.2486	0.894
(1, 8)	0.1657	0.1692	0.979	0.1826	0.1954	0.934
(3, 7)	0.1669	0.1791	0.932	0.1852	0.2069	0.895

At $n = 20$ the model’s order of the fifteen classes coincides with the data’s except for the adjacent pair (1, 6), (3, 5) (which differ by 3% in the data); the model is below the data in every class but (1, 1), by 5–12%, and at $n \approx 30$ by 6–12% in every class but (1, 1) and (1, 2). The model thus reproduces the fifteen measured constants within 12% (model/data 0.89–1.05 at $n = 20$, 0.88–1.02 at $n \approx 30$) and their order at matched n up to one adjacent pair.

Blind test. The first eight classes were known to us when the model was built. The remaining seven were *not*: their model values were written into a dated ledger (3 September 2026, 16:05 WITA) before a second agent, who had not seen the model or the ledger, measured them on the database with independent code; all seven fell within 2–10% of the prediction (model/data at mixed n : 0.93, 0.95, 0.95, 0.90, 0.93, 0.98, 0.94) and in the predicted order up to pairs differing by less than 3%. At matched $n = 20$ the agreement for the long directions improves (model/data 0.93 for both (1, 7) and (3, 7) instead of 0.87 and 0.85 against the mixture).

6.4 What fails, and what is not claimed

A first-order pair model fails. The natural first attempt — a pair of direction v survives with probability $(1 - \rho)^L$, ρ the conditional occupancy of a further cell on its line and L the number of such cells, then renormalise — over-depletes the short directions by a factor of two ($c_{(1,1)} = 0.40$ against 0.73) and puts $(1, 2)$ above $(1, 1)$. The exclusions along a long line are far from independent; the line model succeeds because it treats each line as a unit.

The scale is imposed. The model’s total falls 17% short of (4). After renormalisation the fifteen measured classes sum to 17.1 (model) against 18.6 (data) out of the fixed total $2n - 3 = 37$ at $n = 20$; the model’s deficit in the measured directions is exactly its excess in the tail of rare, steep directions, and no second renormalisation can move it. Of the shortfall, at most a third can be attributed to the row and column sums that the model ignores (a free $2n$ -subset has 705.7 non-axial pairs in expectation, the fixed-marginal model 740); the rest is the model’s treatment of one direction at a time, whereas a solution is constrained in all directions simultaneously.

Drift in n . The model’s values drift by 3–8% between $n = 20$ and 40 (upward for long directions), in the same direction as the data but not by the same amount; the constancy of the short directions in the data is reproduced only approximately ($c_{(1,1)}^{\text{model}}$: 0.764, 0.744, 0.731 at $n = 20, 30, 40$). We found neither a measurement nor an explanation of the growth of the long directions in the literature; the nearest theoretical statement, Remark 1.7 of Ghosal et al. [9], predicts for $k = 2$ an equal contribution of every dyadic scale of the direction length — that is, no growth.

One inversion. At $n = 20$ the model orders $(1, 6)$ above $(3, 5)$ by 0.6%; the data order them the other way by 4.7% (0.2492 against 0.2609) — the model’s largest error, –11.5%, is on $(3, 5)$.

Not claimed. No theorem about the ensemble of solutions is proved here; Theorem 6.1 is a statement about a model. No derivation of the scale, and no explanation of the growth of the long directions with n , is offered.

6.5 Prior art for the spectrum

A multi-channel search (3 September 2026: direct retrieval of the primary sources; full-text search inside arXiv, zbMATH, OEIS, Google Books and Open Library; API queries to Crossref, OpenAlex, Semantic Scholar, GitHub and the Wayback Machine) found no published measurement of pairs per direction, or of diagonal occupation, on actual no-three-in-line solutions; no κ -like constant; no per-direction decomposition of the Guy–Kelly count; and no model of the kind of Section 6.2. The nearest objects, none of which is prior art for the results above, are: Guy and Kelly’s expected number of collinear triples of a random $2n$ -subset [1], a sum over primitive directions that is collapsed through Euler’s function and never split by direction; Flammenkamp’s frequency maps of cell occupation on the same database [3] and his “superficial explanation” of the shading cost of a pair of points, aggregated over symmetry classes; Prellberg’s linear-programming upper bound with capacity-two constraints on rows, columns and the two diagonal families for a parity variant [6] — a relaxation, not an ensemble, and without the weights $\binom{L}{x}$; Simkin’s occupation of diagonals by direction for n -queens (capacity one), with an explicit independence assumption across directions [10]; and a web-posted, unrefereed analysis of July 2026 [12] that measured central directions of symmetric solutions against a random null model and reached the same qualitative conclusion (short directions depleted). In view of the last item the qualitative observation is not claimed as new here; what is new is the quantitative spectrum over all pairs, its fifteen constants at matched n , and the model. *Limits of the search:* free-text web search was not available during it, and several primary sources were not read (Ungar 1982, Jamison 1984–86, Kaplansky–Riordan 1946, Riordan 1958, Hall–Jackson–Sudbery–Wild 1975, Flammenkamp’s JCTA papers of 1992

and 1998, the restricted-occupancy papers of Freund), so the negative result is stated with that caveat; the full record of channels, verdicts and confidence is public in the repository (`docs/research/deep_research_14_answer.md`).

7 Across dimensions

Two of the parts ran the same first-moment test on the other dimension’s data. The results are placed here, where the data live, and are recorded as indications only.

7.1 A first-moment threshold in the cube, refuted by witnesses

Setting $\log \binom{n^3}{m}$ equal to the expected number of coplanar quadruples in a random m -subset of $[n]^3$, computed *as if the constraints were independent*, gives a threshold that landed on exactly $2n + 5$ for $n = 5, \dots, 12$. Against the values then available its slack ran 2, 1, 1, 1, 0, 0: monotonically to zero, which reads as a bound becoming tight. It is not a bound. The witness for $b(11) \geq 28$ exceeds $2n + 5 = 27$.

The failure has a mechanism. The estimate treats the forbidden quadruples as independent; they are positively correlated, since a set that has already avoided one plane avoids the next more easily. A first-moment threshold therefore *understates*, and we verified the sign on the companion estimate as well: the quadruple-independence threshold lies below the certified lower bound at every n from 5 to 12, with the deficit $-1, -2, -3, -3, -4, -4, -5, -4$: growing to $n = 11$, receding by one at the last step. What looked like a bracket around the truth was one biased estimator evaluated at two coarsenings, both leaning the same way.

We record the reading error separately, because it is not specific to this problem: a slack that converges to zero does not distinguish a tight estimate from one about to be crossed, and the two cannot be told apart from inside the model.

7.2 The Guy–Kelly form of the estimate on the cube enumerations

The same test as in Section 5.3 was carried out in the cube (no four coplanar) on the complete enumerations at $n = 3, 4, 5$ of Part I’ by the first solver. There the increment of the error *grows*: +2.22, then +5.53. Two increments *suggest* a wrong exponent — and by our own standard in Section 5.6, where four exact values read as a plateau turned out to lie before the summit, two increments decide nothing. We record this as an indication only. What it does mean is that the conclusion of Section 5.5 is established in the plane and must not be carried into the cube unexamined. This is consistent with the mechanism: in the plane the hard ceiling $2n$ binds and anchors the estimate, while in the cube the corresponding bound $3n$ (Proposition 2.1) stands far from the achievable values and does not.

The two dimensions also share the instrument of Sections 3.4, 4.3 and 4.4: searching inside symmetry strata. In the plane, where complete material exists, the enrichment of maxima by symmetry is measured (factors of 8 to 73 000 at $n = 6, \dots, 11$); in the cube the same enrichment is used, and its price is recorded — the cyclically invariant subspace misses $b(6)$ by one point (Section 4.3), and the gaps of 4, 3, 4, 7 to the 2016 contest records at $n = 17, 19, 23, 29$ measure what confinement to that subspace costs as n grows (Section 4.1).

8 Verification and reproducibility

Witnesses. Every configuration reported as a lower bound in Parts I and I’ was checked by a program sharing no code with the search that found it, and each checker is first given a deliberately corrupted witness that it must reject — a check that cannot fail confirms nothing. For no three collinear the checker tests all $\binom{|S|}{3}$ triples by exact integer cross products and checks

that the points are distinct and inside the cube (`verify_witness_lines.py`). For no four coplanar, configurations were checked twice, by programs written independently and testing different statements: the first computes the 3×3 determinant of every quadruple in exact integer arithmetic (`verify_witness.py`); the second takes every triple, forms the normal of the plane through it, and tests the scalar product against every remaining point, and it also checks that the points are distinct and lie inside the cube, which the first test does not ask. All runs reported zero violations, and the thirteen configurations among the ancillary files were re-verified by the two scripts when this article was assembled.

Certificates and coverage. In Part I the single-call unsatisfiability answers carry DRAT certificates checked by `drat-trim`; the split instances are regenerated deterministically on demand, and their case split, its completeness and its aggregation are each verified separately, by an aggregator that builds the expected pieces from the manifest and refuses a verdict on a missing or undecided piece (Section 3.2). The protocol `docs/VERIFICATION.md` of the repository is written for a reader who trusts none of our code and gives the commands in order.

The plane. Every estimator run behind the tables of Part II is in `logs/kelly/` of the repository, including the measurements that refuted our own earlier claims. For Part II' the measurement code, the model code with an executable statement of Theorem 6.1 checked against brute force for $n \leq 5$, the tests written by a second agent, the two adversarial reviews (one of which found the false claim “the data are constant in n ” in an earlier draft and forced the comparison at matched n), the dated ledger of predictions and the prior-art search of Section 6.5 are public in the repositories below (lemma 005 and need 008 of `lemma-atelier`).

Journals. Every number in this article traces to a file under `logs/` of the repository; by the rule of that repository a number without such a reference counts as unverified, whoever wrote it.

Records. Programs, journals, witnesses, manifests and the verification protocol: <https://github.com/iwasborninbali/saturation>; the papers, packages and witnesses are mirrored at <https://github.com/iwasborninbali/no3-results>; the configurations of Part I with their per-class results are archived in [20]; the lemmas cited in Sections 3.4 and 4.4 are in <https://github.com/iwasborninbali/lemma-atelier>; the database of solutions is Flammenkamp’s [2]. Earlier versions of the four parts were deposited as separate notes at Zenodo, and this article supersedes them:

- Part I, version 1.5: doi:10.5281/zenodo.22273425;
- Part I', version 3.4: doi:10.5281/zenodo.22272371;
- Part II, version 1.9: doi:10.5281/zenodo.22063379;
- Part II', version 1.1: doi:10.5281/zenodo.22278951.

Ancillary files (directory `anc/` of the arXiv package): `verify_witness_lines.py` and the six witnesses of Section 3.4 for $a(8) \geq 94$ (two classes), $a(9) \geq 116$ (two classes), $a(10) \geq 138$ and $a(11) \geq 164$; `verify_witness.py` and the seven configurations of Section 4.4; and, for Part II', `direction_spectrum_model_values.txt` (the model’s constants at $n = 20, 30, 40$ with the factors λ_n) and `direction_spectrum_data_by_strata.txt` (the measured constants by n -stratum, with the solution counts).

9 AI/tool-use disclosure

The searches, the programs and the text of this article were produced by autonomous AI agents (Anthropic Claude) working under the direction of the author of record, who posed the questions, chose what to compute, decided what to claim and takes responsibility for the content. In each part two agents worked with deliberately different program designs and

audited each other’s conclusions rather than each other’s code: one searched, the other verified without reading the searcher’s code, so that the verification would not inherit its blind spots; in Part II one agent worked in each dimension; in Part II’ one built the model and the other measured the database with its own code and did not read the model before the blind test, and a third, adversarial pass reviewed the claims before publication. Several errors reported in the repository were caught that way and not by their author, including one — an unverified symmetry pruning — that no certificate, witness check or coverage check would have detected.

The work is arranged so that this should not matter. Every claim above is meant to be checkable by a third party without trusting the agents or the author: certificates by `drat-trim`, witnesses by brute-force integer arithmetic, coverage by a mechanical aggregator that refuses a verdict on a missing or undecided piece, the encoding by comparison against the direct collinearity criterion, and the model of Part II’ by its dated blind predictions. Where a claim is not checkable in that way, it is listed in the part’s statement of what is not established.

References

- [1] R. K. Guy and P. A. Kelly, The no-three-in-line problem, *Canad. Math. Bull.* 11 (1968) 527–531.
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